

ENSHI, China

WMO: 574470

Lat: 30.2861N

Lon: 109.4711E

Elev: 458

StdP: 95.94

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.1	0.8	-4.5	2.7	4.2	-2.7	3.2	4.3	3.7	5.9	3.2	5.4	1.0	270	0.322

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	35.4	24.7	34.1	24.3	32.8	24.0	26.2	31.6	25.7	30.8	25.3	30.0	2.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	21.4	27.6	24.6	20.7	27.4	24.1	20.1	27.3	84.7	31.9	82.7	31.0	80.8	30.2	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
3.8	3.1	2.5	DB	-1.6	37.4	1.2	0.8	-2.5	37.9	-3.2	38.4	-3.8	38.9	-4.7	39.4	(4)
			WB	-2.2	27.2	1.2	0.6	-3.1	27.6	-3.8	28.0	-4.4	28.4	-5.3	28.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	5.7	7.8	12.3	17.6	21.4	24.7	27.7	27.6	23.5	18.0	12.7	7.3
	DBStd	8.13	2.46	2.83	3.64	3.51	2.74	2.70	2.28	2.56	3.11	2.96	2.92	2.19
	HDD10.0	319	134	71	18	1	0	0	0	0	0	9	86	
	HDD18.3	1490	392	294	191	54	6	0	0	2	43	168	341	
	CDD10.0	2964	2	11	88	228	354	441	548	547	404	248	91	3
	CDD18.3	1093	0	0	2	32	101	192	289	289	156	32	1	0
	CDH23.3	9427	0	0	23	228	629	1453	2913	2956	1081	143	1	0
	CDH26.7	3804	0	0	3	58	198	536	1258	1342	383	27	0	0
Wind	WSAvg	1.1	0.9	1.0	1.2	1.2	1.1	1.1	1.3	1.3	1.2	1.0	0.9	0.9
Precipitation	PrecAvg	1470	26	33	67	128	192	226	244	155	186	119	65	31
	PrecMax	1954	58	78	160	227	339	475	518	433	409	259	153	73
	PrecMin	1026	4	4	25	26	83	93	36	33	22	15	2	3
	PrecStd	242	13	18	34	49	62	103	132	83	98	61	32	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	18.7	26.1	31.0	32.9	35.3	36.7	36.9	34.8	29.7	22.6	15.0
		MCWB	9.5	12.2	16.4	19.2	20.1	23.3	25.4	24.7	24.0	20.5	15.7	10.4
	2%	DB	12.3	15.7	22.9	28.3	30.9	33.2	35.2	35.6	32.5	26.8	20.4	13.1
		MCWB	7.9	10.5	14.7	18.2	20.5	23.1	25.2	24.7	23.1	18.8	14.9	9.8
	5%	DB	10.5	13.5	20.5	26.1	28.9	31.5	33.9	34.3	30.7	24.7	18.5	11.8
		MCWB	7.6	9.7	13.5	17.7	20.0	23.0	24.9	24.4	22.3	18.3	14.3	9.1
	10%	DB	9.2	12.0	18.1	23.6	26.9	29.9	32.5	32.9	28.8	22.7	16.9	10.6
		MCWB	7.1	9.6	12.7	17.0	19.9	22.8	24.8	24.2	21.6	17.9	14.0	8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	13.5	18.2	21.6	23.9	26.1	27.1	26.6	25.5	22.2	17.2	12.6
		MCDB	12.8	16.2	22.9	26.5	27.6	30.5	32.4	33.0	31.0	25.8	19.6	13.4
	2%	WB	9.3	11.9	15.9	20.4	22.7	25.2	26.4	26.0	24.7	20.8	16.2	10.7
		MCDB	10.8	14.0	20.5	24.6	26.4	29.3	31.6	31.9	29.5	24.1	18.2	11.9
	5%	WB	8.2	10.9	14.7	19.1	21.7	24.7	25.9	25.5	24.0	19.6	15.4	9.5
		MCDB	9.8	12.3	18.4	23.1	25.7	28.3	31.1	31.0	28.2	22.6	17.3	10.9
	10%	WB	7.4	9.9	13.6	17.9	20.9	24.0	25.5	25.1	23.2	18.7	14.5	8.6
		MCDB	8.9	11.6	16.9	22.1	24.6	27.3	30.2	30.2	26.8	21.5	16.5	10.1
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.7	7.7	8.4	8.2	7.8	8.2	8.8	7.7	6.4	5.7	4.9
		MCDBR	8.9	10.4	13.4	14.2	13.4	11.9	10.8	11.3	11.4	11.0	9.5	7.8
		MCWBR	4.7	5.2	5.4	4.9	4.2	3.3	2.5	2.4	3.1	3.8	4.3	4.2
	5% WB	MCDBR	6.5	6.8	10.7	10.5	9.0	7.9	9.0	9.4	8.5	8.0	7.2	5.6
		MCWBR	3.8	3.9	4.8	4.5	3.7	2.9	2.5	2.5	2.9	3.4	3.7	3.5
Clear-Sky Solar Irradiance	taub	0.670	0.701	0.732	0.715	0.632	0.621	0.587	0.627	0.634	0.662	0.672	0.643	
	taud	1.490	1.466	1.453	1.528	1.732	1.796	1.903	1.780	1.731	1.607	1.547	1.543	
	Ebn at Noon	539	578	604	643	704	709	732	696	668	600	535	531	
	Edh at Noon	249	277	299	286	234	219	196	219	223	237	233	226	
All-Sky Solar Radiation	RadAvg	1.63	1.88	2.92	3.57	3.75	4.00	4.62	4.49	3.37	2.52	2.01	1.64	
	RadStd	0.35	0.37	0.26	0.49	0.66	0.57	0.62	0.46	0.44	0.43	0.36	0.34	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.34	N/A	N/A	+0.65	N/A	N/A	N/A	-66	+120	+64
Regional (5 neighbors)		N/A	N/A	N/A	+0.65	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page